Two-Column Proofs — Notes — Answer Key

Watch the video and fill in each blank. Pause when you see “Your turn.”

Name: _____ Date: _____

☞ *Video: What a two-column proof looks like*

Key Idea

Statements	Reasons
1. . . . (given)	Given
2. . . .	a property, definition, . . .
. . . (what you must prove)	. . .

Start with what is **given**. End with what you must **prove**. Every line must follow from the lines **above** it.

☞ *Video: An algebra proof*

Watch Me — Worked Example

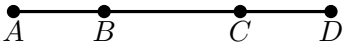
Given: $2x + 5 = 13$ Prove: $x = 4$

Statements	Reasons
1. $2x + 5 = 13$	Given
2. $2x =$ 8	Subtraction Property of Equality
3. $x = 4$	Division Property of Equality

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☞ *Video: A geometry proof*

Watch Me — Worked Example



Given: $AB = CD$ Prove: $AC = BD$

Statements	Reasons
1. $AB = CD$	Given
2. $BC = BC$	Reflexive Property
3. $AB + BC = BC + CD$	Addition Property of Equality
4. $AB + BC = AC,\ BC + CD = BD$	Segment Addition Postulate
5. $AC = BD$	Substitution Property

☞ *Video: Watch out*

Common Mistake

A reason must be a **given**, a **definition**, a **property**, a **postulate** or a **theorem** — never "it looks like it" or "I calculated it".

☞ *Video: Your turn*

Your Turn

Given: $5x - 2 = 18$ Prove: $x = 4$

1. $5x - 2 = 18$	Given
2. $5x = 20$	Addition Property of Equality
3. $x = 4$	Division Property of Equality

Notice

Remember: **Statements** on the left, **reasons** on the right. Start with the **given**; end with what you must **prove**. Every reason is a given, definition, property, postulate or theorem.

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